

Semantic Similarity in Latent Space for LLM Prediction Evaluation

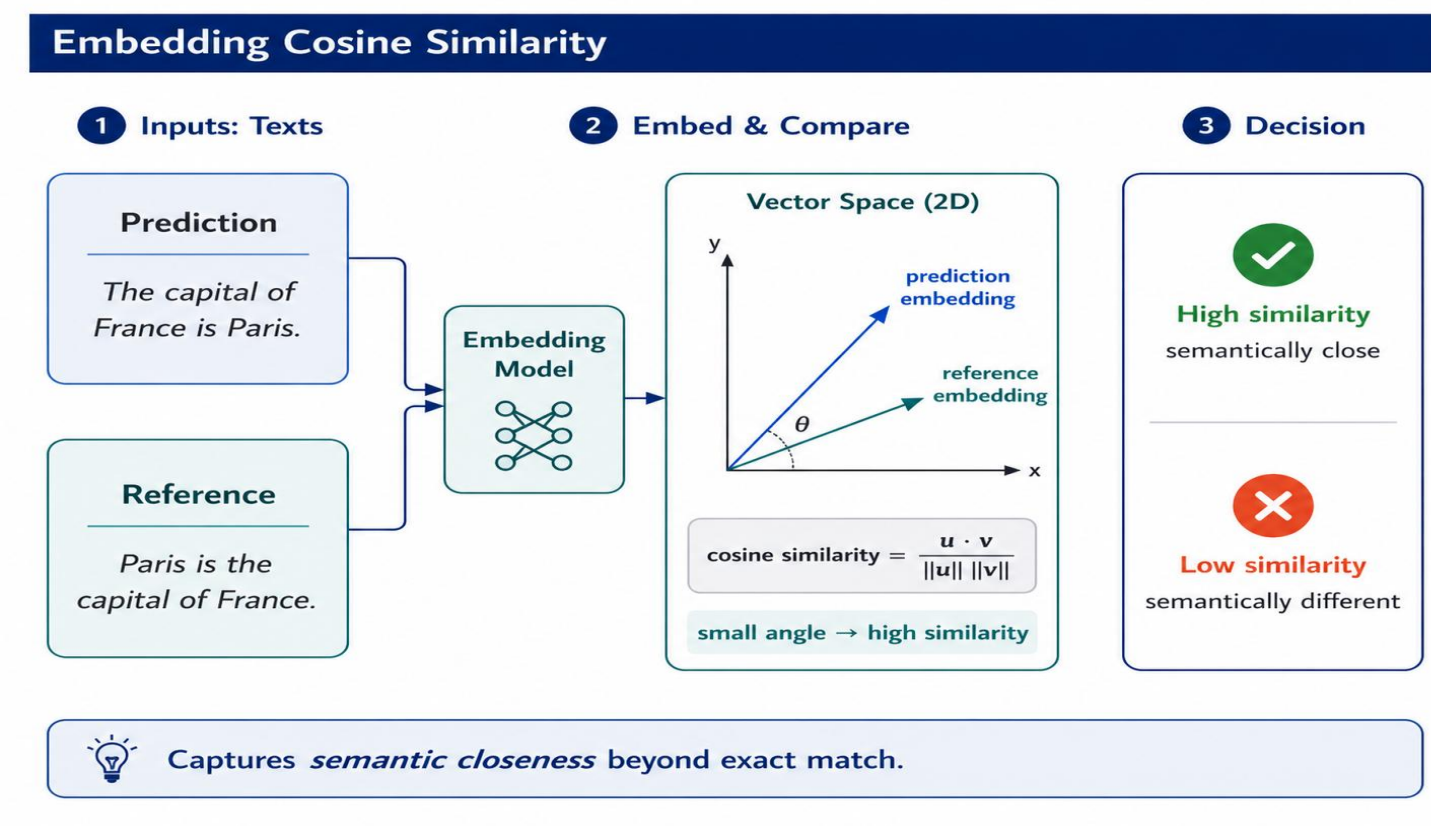
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AIAA 4051 Final Research Project, Introduction to NLP, HKUST(GZ)

Problem Background

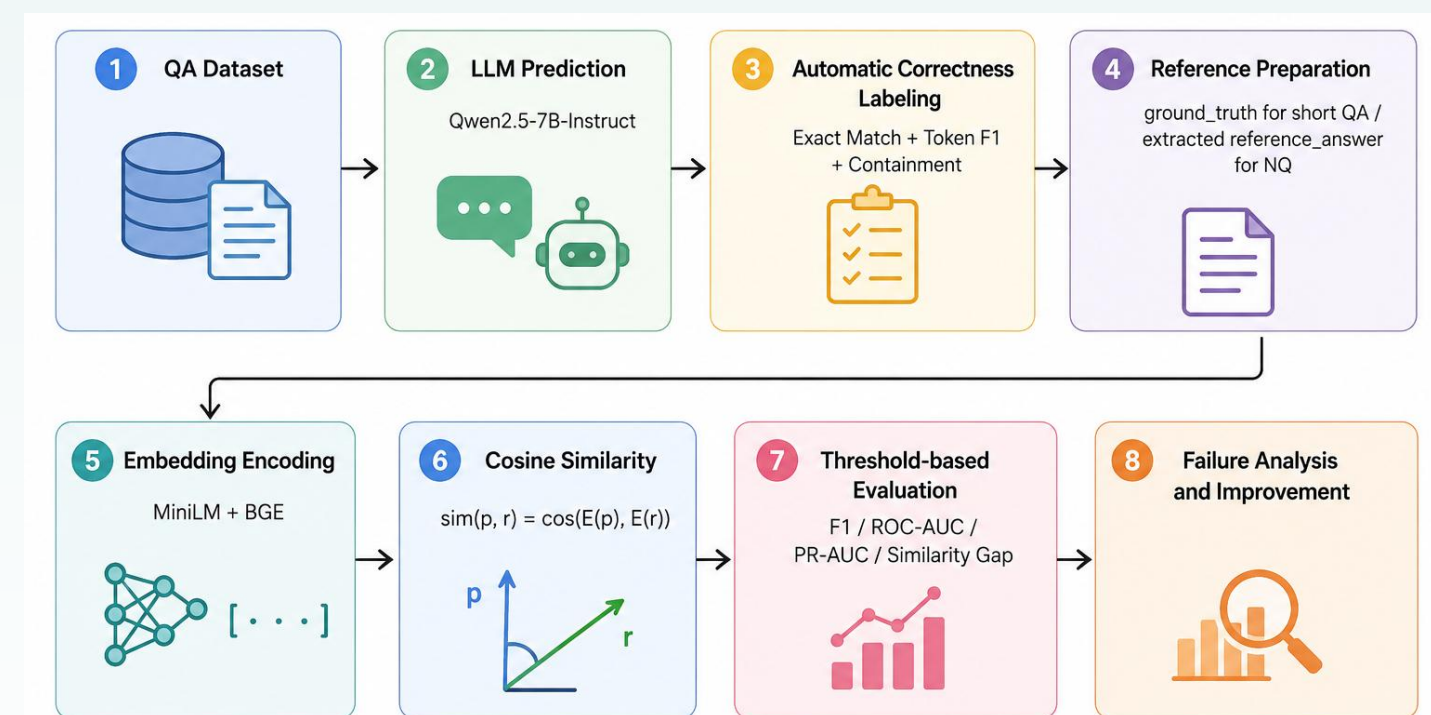
LLMs can produce fluent QA answers, but judging correctness is difficult. Exact-match metrics miss valid paraphrases, while embedding similarity may capture meaning better. However, semantic similarity can confuse topic overlap with factual correctness, especially in long answers.

Research Question:

Can **cosine similarity** between prediction and reference embeddings distinguish correct from incorrect LLM-generated QA answers?



Methodology



The correctness label is **automatically generated** rather than manually annotated. It combines exact match, token-level F1, and containment rules. Therefore, our evaluation measures whether embedding similarity agrees with a reproducible automatic correctness signal, while failure analysis identifies where this signal is noisy or incomplete.

Experiment

Experiment

We evaluate the similarity-based correctness signal on both **short-form** and **long-form** QA datasets. The main experimental variable is reference granularity: short-form datasets usually provide compact answer spans, while NQ originally provides long Wikipedia evidence passages.

Setup

LLM: Qwen2.5-7B-Instruct

Embedding models: **MiniLM** and **BGE**

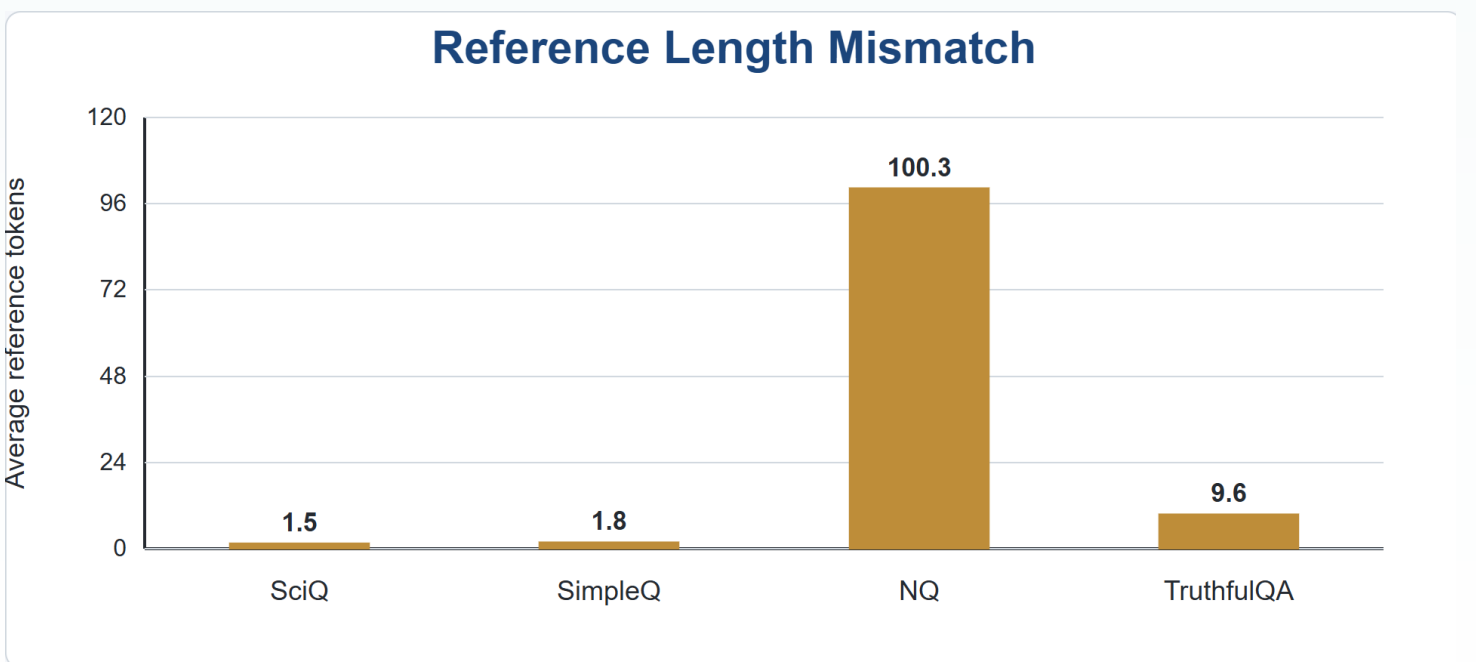
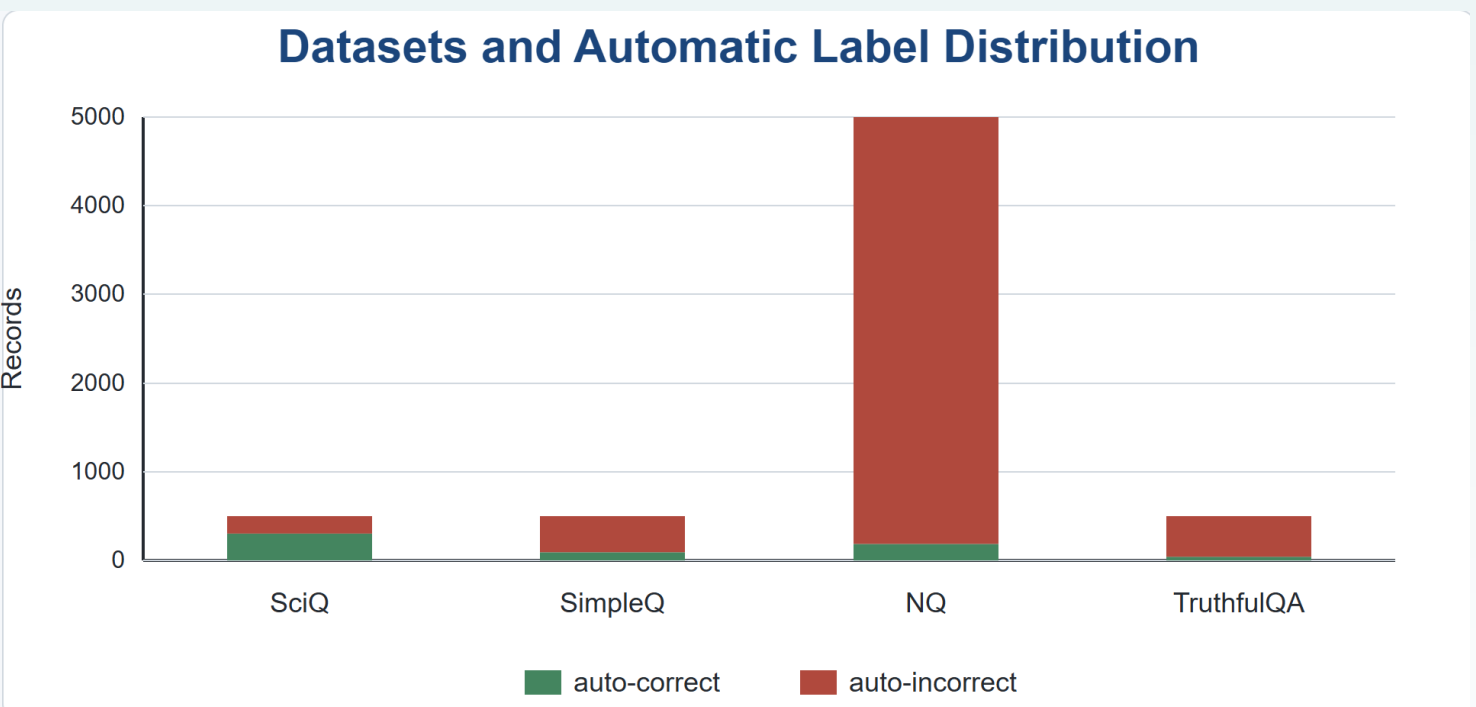
Similarity metric: cosine similarity

Decision rule: similarity $\geq$ threshold $\rightarrow$ predicted correct

Labels: rule-based automatic correct_label

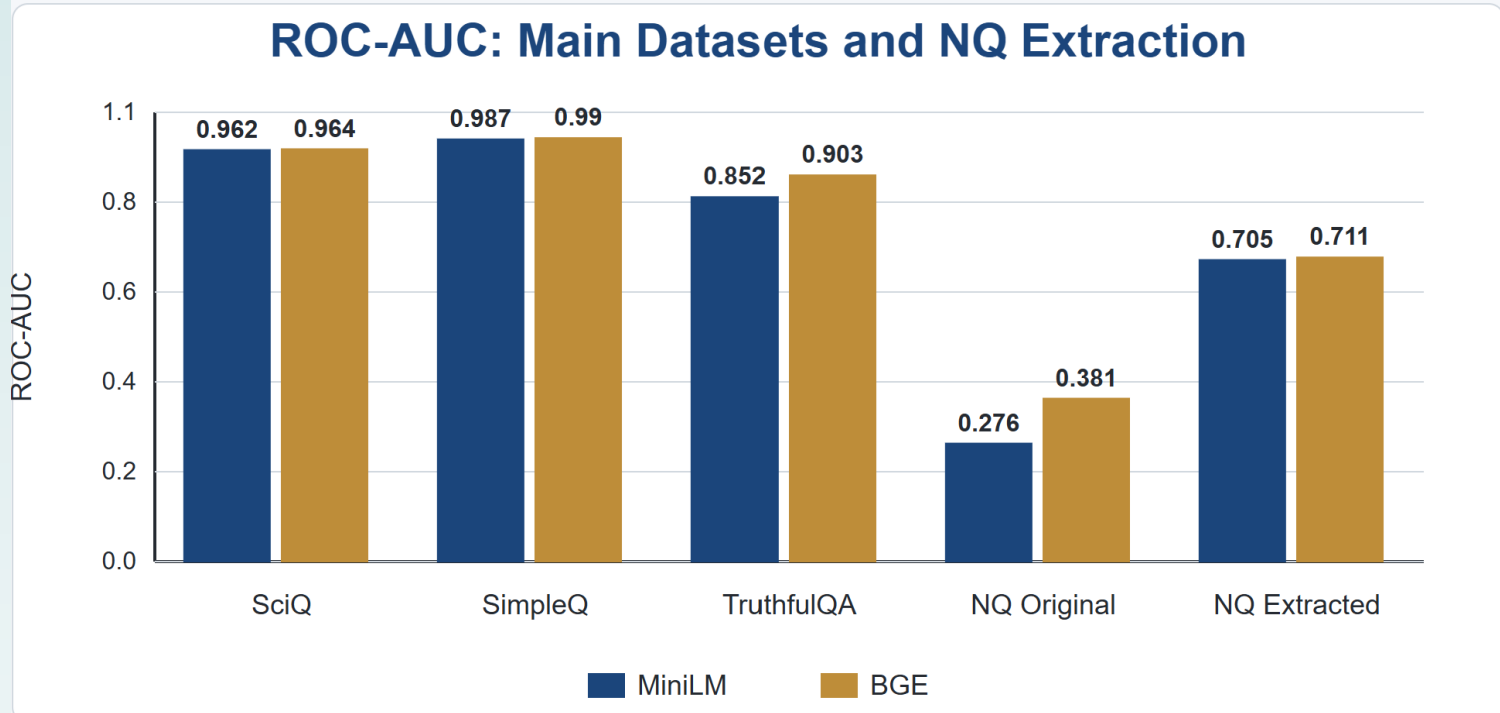
Metric	Purpose
ROC-AUC	Measures ranking quality across all thresholds
Best F1	Finds the strongest threshold-based classifier
Fixed-threshold F1	Tests a simple global threshold
Similarity Gap	Difference between correct and incorrect mean similarity
Failure Counts	Counts high-similarity wrong and low-similarity correct cases

Dataset

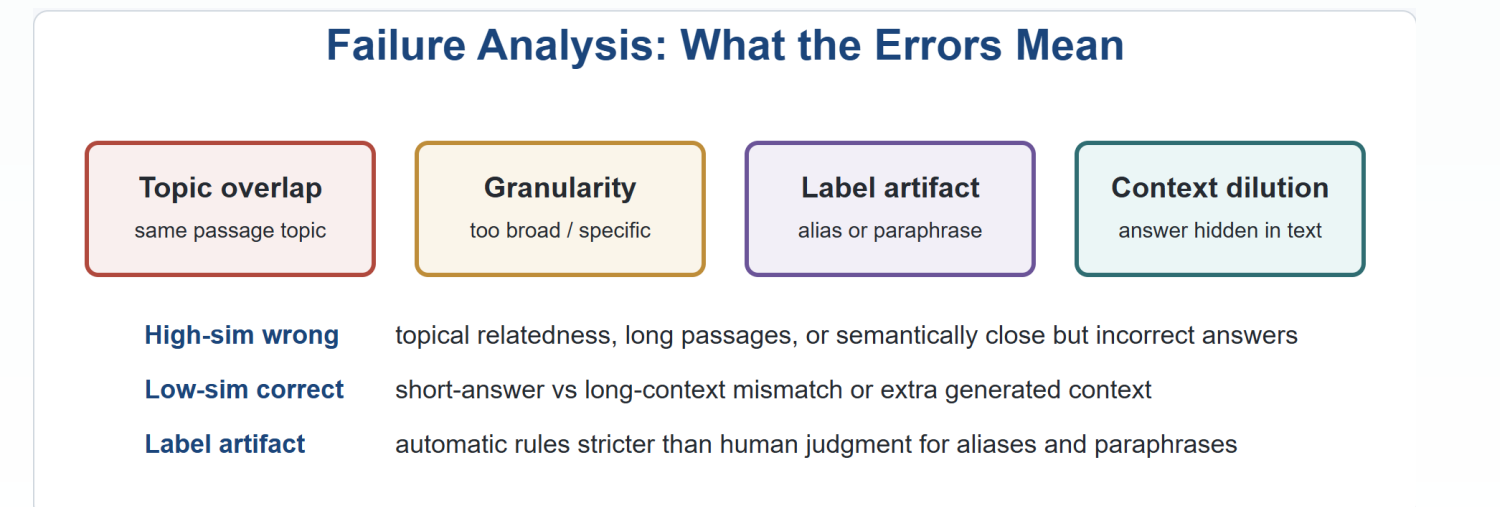
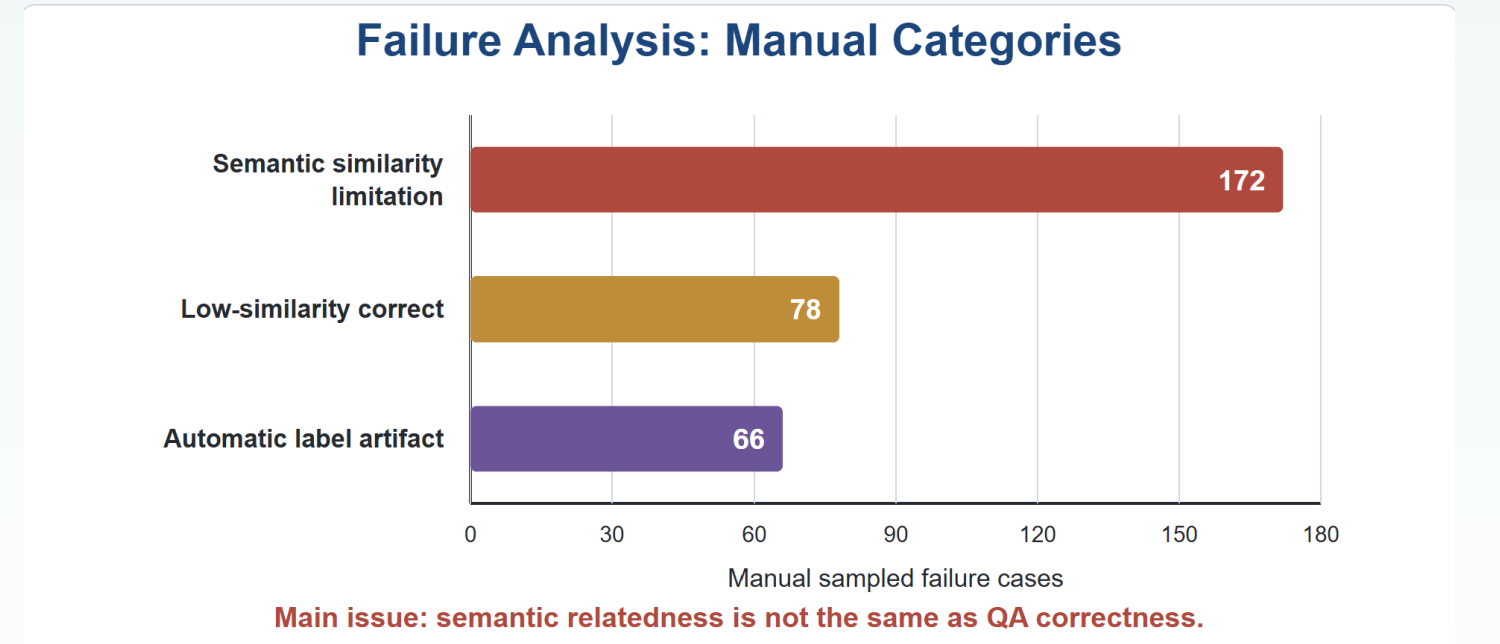


Results

Dataset	Type	MiniLM ROC-AUC	BGE ROC-AUC	Finding
SciQ	Short-form	0.962	0.964	Strong signal
SimpleQ-Wiki	Short-form	0.987	0.990	Strongest signal
TruthfulQA	Long-form	0.852	0.903	Useful but harder
NQ Original	Long-form	0.276	0.381	Fails under long-reference mismatch



- Short-form QA shows **strong separation** between correct and incorrect predictions.
- TruthfulQA remains positive but is harder due to open-ended factual statements.
- Original NQ fails because short answers are compared with **long evidence passages**.



Advanced

1. Baselines & Ablation Studies

We treat every improvement as an ablation. Starting from sentence-level embedding cosine, we compare lexical baselines, entity/token overlap, embedding-overlap hybrids, NQ reference extraction, prediction-span similarity, factual conflict penalties, and reduced multi-view hybrid scores.

Key results:

- NQ reference_answer extraction improves ROC-AUC from **0.269/0.391** to **0.705/0.711**.
- Span-level similarity greatly improves ranking, with best F1 around **0.80**.
- Factual conflict penalties reduce high-similarity wrong cases caused by number/date/entity mismatch.
- Unit 6 span-guarded multi-view hybrid gives the best fixed-threshold performance: **MiniLM F1 = 0.725**, **BGE F1 = 0.701**.

2. Theory: Multi-View Embedding Evaluator

Core Idea

Correctness $\neq$ semantic similarity only

Correctness = semantic match + lexical evidence + factual-unit consistency + answer granularity + question-type awareness

This keeps embedding similarity as one useful signal, but prevents it from becoming the only judge of factual correctness.

3. Case Studies

Case Study1

Field	Content
Question	Which river is the water source for Victoria Falls?
Prediction	Zambezi River
Original Reference	Long passage about Victoria Falls
Problem	Correct short answer is compared with a long evidence passage
Fix	Extract reference_answer = Zambezi River

Case Study2

Field	Content
Question	Who owns Aquarius Casino in Laughlin, NV?
Reference	Golden Entertainment
Prediction	Caesars Entertainment
Problem	Same topic, wrong entity
Needed View	Entity consistency / verifier